Technical Report: Shipment Route Optimization

Problem and objective

Logistics planners balance shipment cost, transit time, and reliability. This project is an explainable decision-support API that evaluates alternative routes and recommends the best route for stated priorities. It is a demonstration, not an operational routing engine.

Dataset and system design

The synthetic dataset contains 20 Indian origin-destination shipments and 60 route options. Each option records hubs, INR cost, transit days, delay probability, congestion risk, and geopolitical risk. CSV/Pandas load the data into SQLite; FastAPI exposes it; independent scoring, Pareto, and explanation modules make the logic testable.

Decision framework and explainability

Combined risk = 0.50 x delay probability + 0.30 x congestion risk + 0.20 x geopolitical risk. Every risk input is a 0-1 estimate. Delay gets the highest internal weight because it most directly affects delivery reliability.

For each shipment, cost, transit time, and combined risk are min-max normalized, then inverted because lower is preferable: benefit = 1 - (value - minimum) / (maximum - minimum). Equal metric values receive benefit 1.

The default final score is 0.40 x cost benefit + 0.30 x time benefit + 0.30 x risk benefit. Users can submit non-negative weights, normalized proportionally. The response ranks every route and explains the winner against the runner-up, making trade-offs visible rather than hiding them behind a score.

Multi-objective evaluation, limitations, and next steps

Pareto analysis minimizes cost, time, and combined risk independently. A route is excluded when another option is no worse on all objectives and better on at least one. The remaining non-dominated routes are valid alternatives: for example, a cheaper route may be slower than a more reliable air option.

Automated tests verify the risk formula, priority-sensitive ranking, Pareto dominance, successful API responses, and invalid input handling. This evaluates consistency of the decision framework and demonstrates sensitivity to business priorities.

All values are illustrative and static. A production version should use live carrier quotes, weather, traffic, port congestion, capacity, carbon constraints, calibrated risk estimates, audit history, authentication, and monitoring.